

Computer Organization & Assembly Language

Lab 04

Lab Instructor:

Mr. Tariq Mehmood Butt tariq.butt@pucit.edu.pk

Teacher Assistants:

Hafiz Muhammad Ahmad	bcsf21m502@pucit.edu.pk
Muhammad Umair	bsdsf23m006@pucit.edu.pk
Muhammad Zohaib	bsdsf23m023@pucit.edu.pk

Lab Objective:

To understand how to write, assemble, link and execute an assembly program using .ASM files.

Instructions:

You are required to prepare the .ASM file of each task and run the executable file in the debugger. For each Trace call, carefully observe and note the changes in registers.

Note: You are not required to handle the invalid inputs.

Lab Tasks

Task 1: Define two 8-byte arrays, `array1` and `array2`. Write an assembly code to copy the content of `array1` to `array2` and display the content of both arrays after copying.

Task 2: Define two 8-byte arrays, `array1` and `array2`. Write an assembly code to copy the content of `array1` to `array2` in reverse order and display the content of both arrays after copying.

Task 3: Define two 8-byte arrays, `array1` and `array2`. Write an assembly code to swap the contents of `array1` with `array2` and display the content of both arrays after swapping.

Task 4: Define two 8-byte arrays, `array1` and `array2`. Write an assembly code to swap the contents of `array1` with `array2` in reverse order and display the content of both arrays after swapping.

Task 5: Write an assembly program that takes a character input from user and displays it.

Task 6: Write an assembly program that takes an alphabet character input and tells whether it is capital or small. Display “Capital” for capital letters and “Small” for small letters.

Task 7: Write an assembly code which takes two single-digit numbers as input and adds them. Display the correct output if the sum is less than 10; otherwise, display “Overflow”.

Task 8: Write an assembly program that takes a single character as input and checks whether it is a digit (0–9) or not. Display “Digit” if true, otherwise display “Not a Digit.”